Internet Crimes and Cyber Security

Week: 05 **Session: 23**

Topic : **Tools and Techniques for Penetration Testing**

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Tools and techniques to Counter Cyber Attacks

4. Countering Cyber Attacks technically using Penetration Testing Tools

Penetration testing types are broadly categorized by knowledge level (Black Box, White Box, Grey Box) or by the target system (Network, Web App, Cloud, Mobile, API, IoT), all aiming to find and fix vulnerabilities by simulating real attacks, from zero-knowledge external assaults to insider threats. Black Box Testing: Testers have zero knowledge, simulating external hackers trying to breach systems blindly.

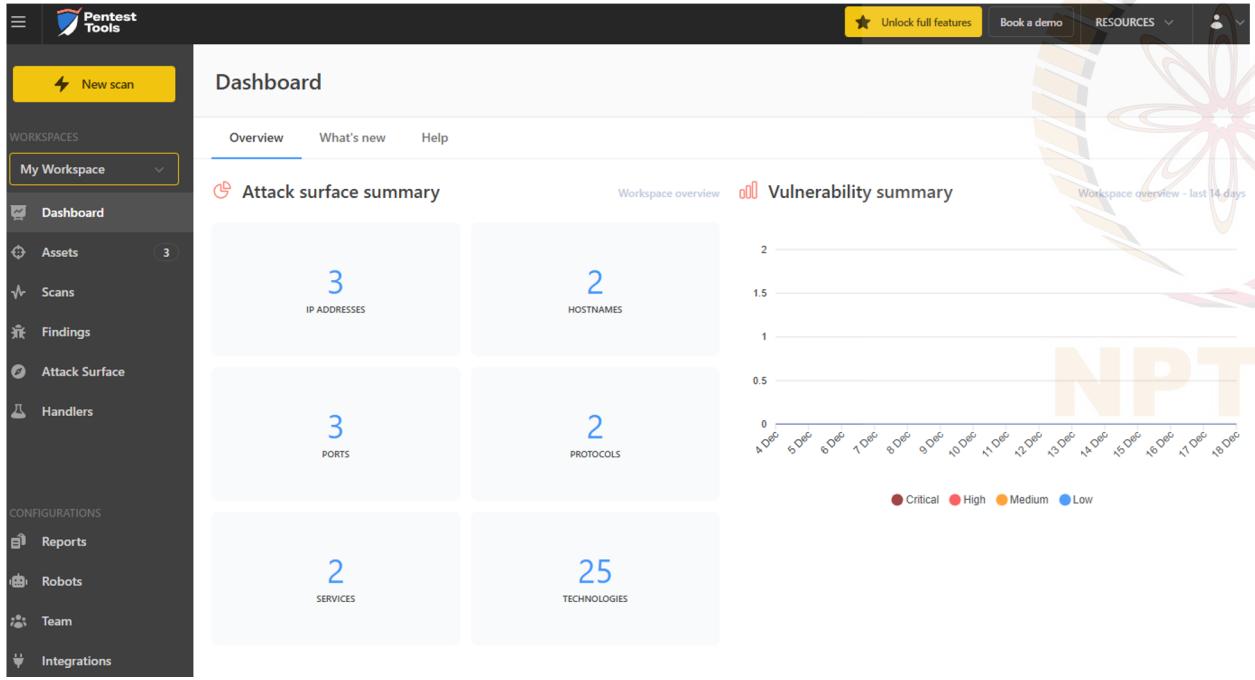
1. **White Box Testing**: Testers have full access (code, network diagrams, credentials), allowing deep, comprehensive analysis.
2. **Grey Box Testing**: Testers have partial knowledge (e.g., user login), mimicking insider threats or attackers with some initial access.

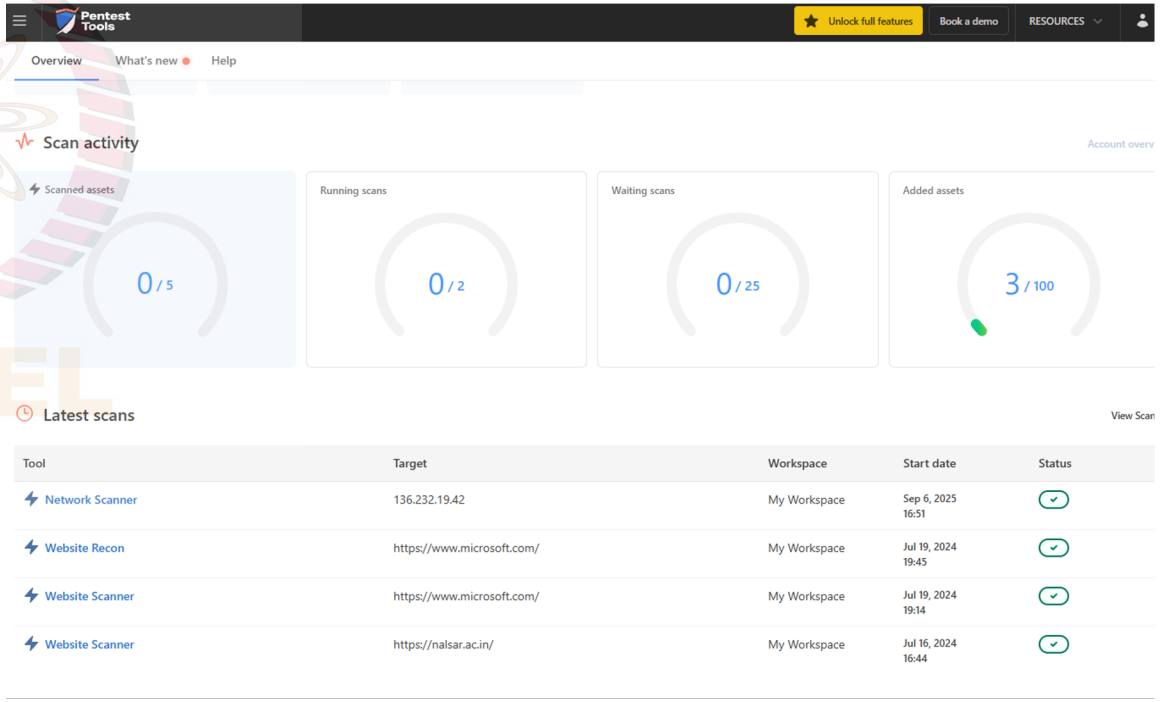
By Target/System

1. Network Penetration Testing: Focuses on internal/external network infrastructure (firewalls, routers).
2. Web Application Testing: Finds flaws in web apps, including APIs and web services.
3. Cloud Penetration Testing: Assesses security of cloud environments (AWS, Azure, GCP).
4. Mobile Application Testing: Checks security of iOS/Android apps and their backend.
5. API Penetration Testing: Targets Application Programming Interfaces for vulnerabilities.
6. IoT Penetration Testing: Examines security of Internet of Things devices.
7. Wireless Penetration Testing: Assesses Wi-Fi network security.

Tools and techniques to Counter Cyber Attacks

Penetration Testing Tool : https://pentest-tools.com/





Tools and techniques to Counter Cyber Attacks

5. Understanding Incidence Response and Recovery Platforms

Incident Response Process

Preparation	• Processes, Procedures, and Resources • Create an incident response plan, define roles and duties, identify important employees, and defining communication routes
Detection & Analysis	• Identify a security event through intrusion detection systems, log monitoring, network monitoring, or user reporting • Investigate the incident and identify the nature and severity of the issue
Containment & Mitigation	• Isolating and stopping impacted systems or taking other appropriate measures to limit the impact
Investigation & Forensics	• Investigate to determine the cause, scope, and extent of the occurrence. • Evidence gathering - Forensic, Log Analysis, Detemine Access point of Vulnerabilities exploited
Communication & Reporting	• Communication is critical throughout the incident response process. • Include Management, IT teams, legal counsel, and, law enforcement authorities in communication
Recovery	• After the event has been controlled and investigated, focus your attention on restoring regular operations
Lessons Learnt & Protection	• Put proactive steps in place to strengthen the defenses, • Improve incident response skills, and defend against similar occurrences in the future

Thank You